

SOLUTIONS

1. C is correct. If Scenario 1 occurs, the expected recovery is 60% (USD50,000) + 40% (USD30,000) = USD42,000, and if Scenario 2 occurs, the expected recovery is 90% (USD80,000) + 10% (USD60,000) = USD78,000. Weighting by the probability of each scenario, the expected recovery is 40% (USD42,000) + 60% (USD78,000) = USD63,600. Alternatively, first calculating the probability of each amount occurring, the expected recovery is (40%)(60%)(USD50,000) + (40%)(40%)(USD30,000) + (60%)(90%)(USD80,000) + (60%)(10%)(USD60,000) = USD63,600.
2. A is correct. The analyst must first calculate expected sales as $0.05 \times \text{USD}70 + 0.70 \times \text{USD}40 + 0.25 \times \text{USD}25 = \text{USD}3.50 \text{ million} + \text{USD}28.00 \text{ million} + \text{USD}6.25 \text{ million} = \text{USD}37.75 \text{ million}$.

After calculating expected sales, we can calculate the variance of sales:

$$\sigma^2 (\text{Sales})$$

$$= P(\text{USD}70)[\text{USD}70 - E(\text{Sales})]^2 + P(\text{USD}40)[\text{USD}40 - E(\text{Sales})]^2 + P(\text{USD}25)[\text{USD}25 - E(\text{Sales})]^2$$

$$= 0.05(\text{USD}70 - 37.75)^2 + 0.70(\text{USD}40 - 37.75)^2 + 0.25(\text{USD}25 - 37.75)^2$$

$$= \text{USD}52.00 \text{ million} + \text{USD}3.54 \text{ million} + \text{USD}40.64 \text{ million}$$

$$= \text{USD}96.18 \text{ million.}$$

The standard deviation of sales is thus $\sigma = (\text{USD}96.18)^{1/2} = \text{USD}9.81 \text{ million}$.